MIL-S-15769C
26 June 1970
SUPERSEDING
MIL-S-15769B
1 September 1964

MILITARY SPECIFICATION

STRAIGHTEDGES, STEEL (MACHINISTS' AND TOOLMAKERS')

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers machinists' and toolmakers' straightedges for general machine shops.

★ 1.2 Classification.- Straightedges shall be of the following types and length, as specified (see 6.2 and 6.3).

Type I	-	Machinists', square edges, not graduated, semi-hard
Type II	-	Machinists', one edge beveled, one edge square, not graduated, semi-hard
Type V	-	Tool makers' knife edge, not graduated, hardened
Class 1	-	Individual straightedge
Class 2	-	Straightedge set
Lengths	-	See tables I through III

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

FSC 5210

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STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18-67 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Standard product.- The items covered by this specification shall be the manufacturer's current, standard commercial product except for any changes necessary to comply with specification requirements.

* 3.2 Materials.- Material for the straightedges shall be tool steel of the quality normally used by the supplier for steel straightedges, provided the completed item complies with all provisions of this specification.

3.3 Construction.-

3.3.1 Types I and II straightedges.- All surfaces of the types I and II straightedges shall be flat and parallel with the opposite surface. Except for one edge of type II straightedge, the ends and working edges of each straightedge shall be squared (90 degrees ± 5 minutes) for the full thickness. One edge of the type II straightedge shall be uniformly beveled and squared for the full width of the land (see 4.3.1).

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* 3.3.2 Type V straightedge.- All working edges shall be straight and parallel with one edge beveled to a knife edge and one or more ends squared (see 4.3.1). Class 1 straightedge shall be furnished with a case in accordance with manufacturers' commercial practice. Class 2 shall consist of a case containing a glass test bar and one each of all items listed in table III.

3.3.3 Steel.- Straightedges shall be made of tool steel and shall have a Rockwell (C-scale) hardness as specified in tables I to III, when tested as specified in 4.3.2.

3.3.4 Dimensions and tolerances.- Straightedges shall be of the dimensions and within tolerances specified in tables I to III, inclusive. The width and thickness dimensions shall each be uniform throughout, except for type V straightedges, which may have one end beveled.

3.4 Support points.- Straightedges, 36 inches in length or longer, shall have two support points stamped or otherwise permanently marked on at least one surface (see table II for dimensional locations of support points.)

3.5 Marking.- The straightedges shall be marked for identification with the manufacturer's name or readily identifiable trademark.

3.6 Finish.- All the surfaces of straightedges shall be ground and polished, except the working edges shall be ground and lapped.

3.7 Workmanship.- All surfaces of straightedges shall be smooth with no burr, sharp edges, cracks, gouge, grind chatter, or burn marks.

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TABLE I.- Type I - Machinists', square edges, not graduated, semi-hard

Length ± 1/32	Width (approx.)	Thickness (approx.)	Parallelism tolerance for working edges	Straightness tolerance for working edges	Hardness Rockwell (C)
Inches	Inches	Inch	Inch	Inch	
12	1 - 1-1/2	5/64 - 3/16	0.0005	0.0002	22 - 32
24	1-1/2 - 1-3/4	5/64 - 3/16	0.0008	0.0004	22 - 32

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TABLE II.- Type II - Machinists', one edge beveled, one edge square, not graduated, semi-hard

Length ± 1/32	Width (approx.)	Thickness (approx.)	Parallelism tolerance for working edges	Straightness tolerance on beveled working edge	Straightness tolerance on squared working edge	Hardness Rockwell (C)	Bevel (approx.)	Width land (approx.)	Postion of support point from each end ± 1/64
Inches	Inches	Inch	Inch	Inch	Inch		Degrees	Inch	Inches
36	2 - 3	1/4 - 5/16	0.001	0.0005	0.0006	22 - 32	20	3/64	8
48	2-1/2 - 3	1/4 - 5/16	0.0012	0.0006	0.0007	22 - 32	20	1/16	10-21/32
72	3 - 3-1/4	1/4 - 9/32	0.002	0.0007	0.001	22 - 32	20	5/64	16
96	3-1/4 - 3-1/2	5/16 - 3/8	0.003	0.0009	0.0012	22 - 32	20	5/64	24
120	3-1/2 - 4	5/16 - 7/16	0.004	0.0012	0.0015	22 - 32	20	5/64	30

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TABLE III.- Type V - Toolmakers', knife edge, not graduated, hardened

Length ± 1/32	Width ± 1/64	Parallelism tolerance for working edges	Straightness tolerance on working edges	Hardness Rockwell (C)
Inches	Inch	Inch	Inch	
2-1/4	13/16	0.0001	0.0001	62 - 65
3-1/4	13/16	0.0002	0.0001	62 - 65
4-1/2	13/16	0.0003	0.0001	62 - 65
6-1/4	13/16	0.0004	0.0001	62 - 65

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1, components and materials shall be tested and inspected in accordance with all the requirements of referenced specifications and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection.- The lot shall be all the straightedges of one type and length offered for inspection at one time. The sample unit for this inspection shall be one completely finished straightedge.

4.2.2.1 Visual examination.- The straightedges shall be examined for defects classified in table IV. The inspection level shall be level II with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units.

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TABLE IV.- Classification of defects

Examine	Defect
Finish	Surfaces not ground and polished or ground and lapped, as applicable (see 3.6).
Construction and workmanship	Any support point marking missing (straightedges 36 in. long or longer). Surfaces with sharp edges, cracks, gouge, grind chatter, or burn marks. Case missing. Glass test bar missing (type V).
Identification marking	Missing, illegible, incorrect, incomplete.

* 4.2.2.2 Dimensional examination.- Inspection of the end item shall be made at 68 F. for compliance with the dimensions specified herein. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be S-3 with an AQL of 2.5, expressed in terms of defects per hundred units.

4.2.2.3 End item testing.- The straightedges shall be tested as specified in 4.3.1 and 4.3.2. The inspection level shall be S-1 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.3 Examination of preparation for delivery.- Examination for preparation for delivery shall be made in accordance with PPP-P-40.

4.3 Tests.-

4.3.1 Squareness, flatness and parallelism.- Utilizing standard guages, indicators, and toolmakers' surface plates, the straightedges shall be tested for squareness and flatness of surfaces and for the parallelism of opposite surfaces, as specified in 3.3.1 and 3.3.2. Any nonconformance shall constitute failure of this test.

4.3.2 Hardness.- The straightedges shall be test for hardness at the center and 1 inch from each end, in accordance with ASTM E 18-67. Any hardness reading not within the range for applicable straightedges listed in tables I through III shall constitute failure of this test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.- Preservation, packaging, packing and marking for level A, B or C shall be in accordance with FPP-P-40 (see 6.2).

6. NOTES

6.1 Intended use.- The steel straightedges are intended for use in general shop practice. Types I and II straightedges are intended to be used by machinists to lay out lines and check surfaces on ordinary shop jobs. Type V straightedges are used primarily by toolmakers on work requiring extreme accuracy.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, length, and class of straightedge required (see 1.2).
(c) Levels of preservation, packaging, packing, and marking required (see 5.1).

6.3 Deleted items.- The following straightedges appearing in MIL-S-15769B have been deleted in this revision; since there are no user requirements for the items: -

Type I - 6, 18, 36, 48, 60, and 72 inch lengths

Type II - 12, 18, 24, and 60 inch lengths

Type III - All items

Type IV - All items

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 5210-0201

User activities:

Army - ME, WC, AV
Navy - SH, MC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Straightedges, Steel (Machinists' And Toolmakers') MIL-S-15769C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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